

List of publications by Jorge Alda

Note: All citation figures are given by Inspire-HEP as of September 2, 2026, and exclude self-citations.

1 Published articles

J. Alda, J. Guasch and S. Peñaranda: *Some results on Lepton Flavour Universality Violation*

Eur.Phys. J. C, 79 7 (2019) 588
doi:10.1140/epjc/s10052-019-7092-x
arXiv:1805.03636 [hep-ph]
15 citations.

J. Alda, J. Guasch and S. Peñaranda: *Anomalies in B decays: A phenomenological approach*

Eur.Phys. J. Plus 137 (2022) 217
doi:10.1140/epjp/s13360-022-02405-3
arXiv:2012.14799 [hep-ph]
14 citations.

J. Alda, J. Guasch and S. Peñaranda: *Using Machine Learning techniques in phenomenological studies in flavour physics*

J. High Energ. Phys. 2022, 115 (2022)
doi:10.1007/JHEP07(2022)115
arXiv:2109.07405 [he-ph]
4 citations.

J. Alda, A. W. M. Guerrero, S. Peñaranda and S. Rigolin: *Leptonic Meson Decays into Invisible ALP*

Nucl.Phys.B 979 (2022) 115791
doi:10.1016/j.nuclphysb.2022.115791
arXiv:2111.02536 [hep-ph]
21 citations.

J. Alda, G. Levati, P. Paradisi, S. Rigolin, N. Selimović: *Collider and astrophysical signatures of light scalars with enhanced τ couplings*

JHEP 06 (2025) 008
doi:10.1007/JHEP06(2025)008
arXiv:2407.18296 [hep-ph]
26 citations.

S. Peñaranda, J. Alda, A. Mir: *Flavour Anomalies: A comparative Analysis using a Machine Learning Algorithm*

Int.J.Theor.Phys. 65 (2026) 2, 46
doi:10.1007/s10773-026-06263-y

arXiv:2412.15830 [hep-ph]
2 citations.

J. Alda, C. Brogгинi, G. Di Carlo, L. Di Luzio, D. Piatti, S. Rigolin, C. Toni: *Weak nuclear decays deep-underground as a probe of axion dark matter*
Phys.Rev.D 111 (2025) 3, 035022
doi:10.1103/PhysRevD.111.035022
arXiv:2412.20932 [hep-ph]
15 citations.

J. Alda, J. Asorey, A. Mir, S. Peñaranda: *Physically Consistent Parameter Inference: Transparent Machine Learning Emulation in High Energy Physics and Cosmology*
Universe. 2026; 12(8):238
doi:10.3390/universe12080238
arXiv:2607.12726 [hep-ph]
Invited contribution to special issue “Advances in Machine Learning Techniques in Cosmology and Particle Physics”.
1 citation.

2 Pre-prints

J. Alda, M. Fuentes Zamoro, L. Merlo, X. Ponce Díaz, S. Rigolin: *Comprehensive ALP Searches in Meson Decays*
arXiv:2507.19578 [hep-ph]
Submitted to Fortschritte der Physik.
16 citations.

J. Alda, M. Fuentes Zamoro, L. Merlo, X. Ponce Díaz, S. Rigolin: *ALPaca: The ALP Automatic Computing Algorithm*
arXiv:2508.08354 [hep-ph]
Submitted to Computer Physics Communications.
5 citations.

3 Proceedings

J. Alda, J. Guasch and S. Peñaranda: *Anomalies in B decays: Present status and future collider prospects*
arXiv:2105.05095 [hep-ph]
SLAC eConf C21-03-15.1.
Proceedings to LCWS 2021.
4 citations.

J. Alda, J. Guasch and S. Peñaranda: *Exploring B-physics anomalies at colliders*
arXiv:2110.12240 [hep-ph]
doi:10.22323/1.398.0494

Proceedings to EPS-HEP2021.
1 citation.

A. Mir, J. Alda, S. Peñaranda: *B-Meson Anomalies: Effective Field Theory Meets Machine Learning*

doi:10.22323/1.485.0349

arXiv:2510.17742 [hep-ph]

Proceedings to EPS-HEP2025.

M. Fuentes Zamoro, J. Alda, L. Merlo, X. Ponce Díaz, S. Rigolin: *Tackling ALP searches in meson decays with ALPaca: a phenomenological approach*

doi:10.22323/1.507.0057

Proceedings to 3rd Training School of the COST Action: Cosmic WISPers (CA21106).

4 Lecture Notes

J. Alda: *Lecture notes on Machine Learning applications for global fits*

arXiv:2607.12726 [hep-ph]

Accepted for publication in SciPost Physics Lecture Notes,

<https://scipost.org/submissions/2604.07520v2/>.